
Driver adversary: held-out drivers become indistinguishable in feature space, while distraction classes stay separable (from-scratch ResNet-18)
epoch 1 (early)
epoch 7
epoch 15 (final)
final epoch, colored by CLASS
Without adversary
With adversary
silhouette driver +0.04 | class +0.06
silhouette driver +0.01 | class +0.11
held-out driver (cols 1-3)
p015
p024
p026
p042
p056
p072
class (col 4)
c0 safe driving
c1 texting - right
c2 talking on the phone - right
c3 texting - left
c4 talking on the phone - left
c5 operating the radio
c6 drinking
c7 reaching behind
c8 hair and makeup
c9 talking to passenger
(a) Identity recoverable from features (held-out drivers)
driver-ID probe accuracy
Without adversary
With adversary
chance
epoch
(b) Adversary is pushed to chance (with adversary only)
driver cross-entropy
chance = ln K
driver-head loss (train)
reversal strength λ
λ
epoch
(c) Task performance retained (held-out drivers)
class accuracy (held-out)
epoch
(d) Final: identity removed, task kept
0.73
0.82
0.49
0.46
WHO is driving (probe)
WHAT they do (classifier)